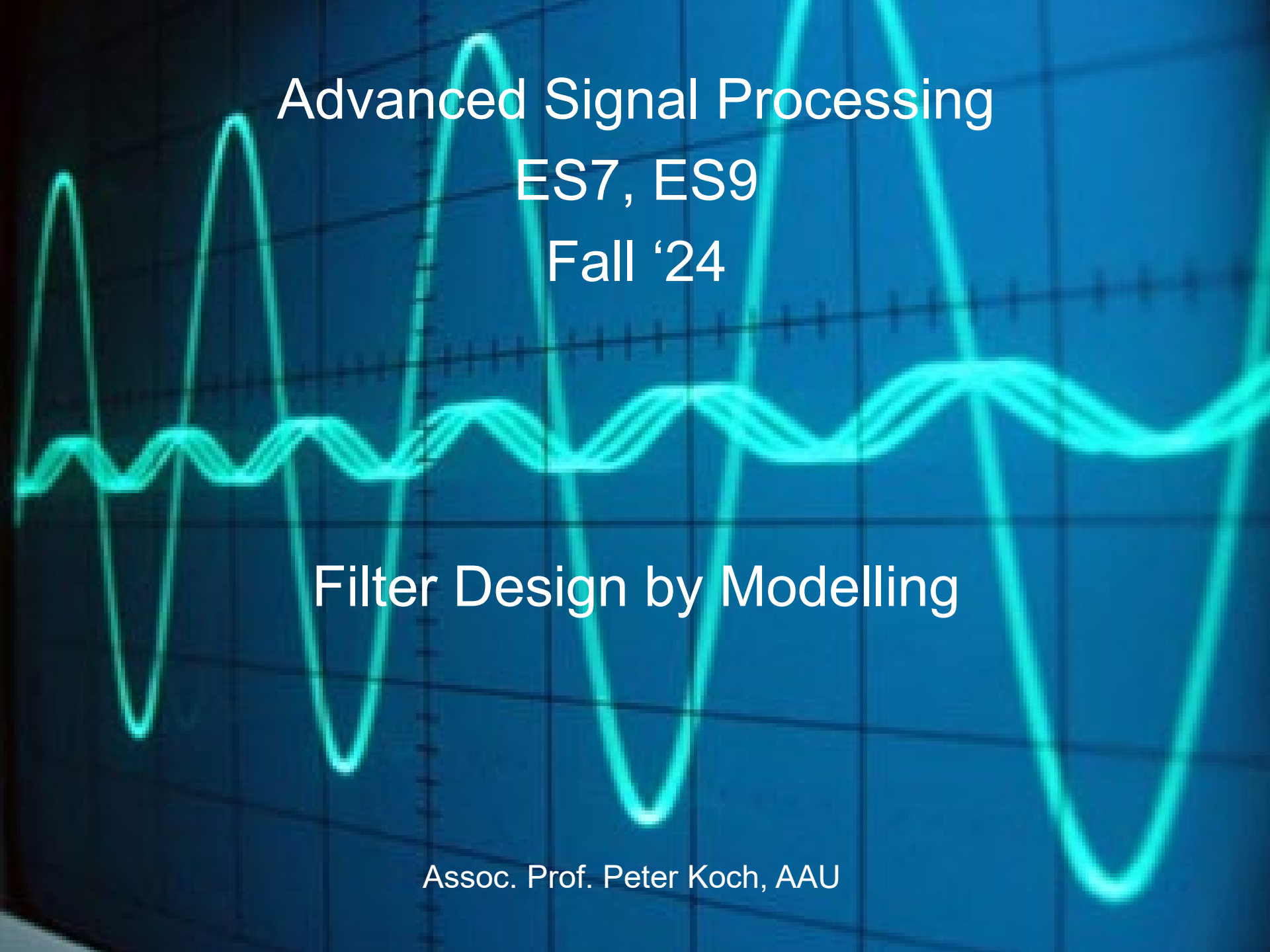




# Advanced Signal Processing

ES7, ES9

Fall '24

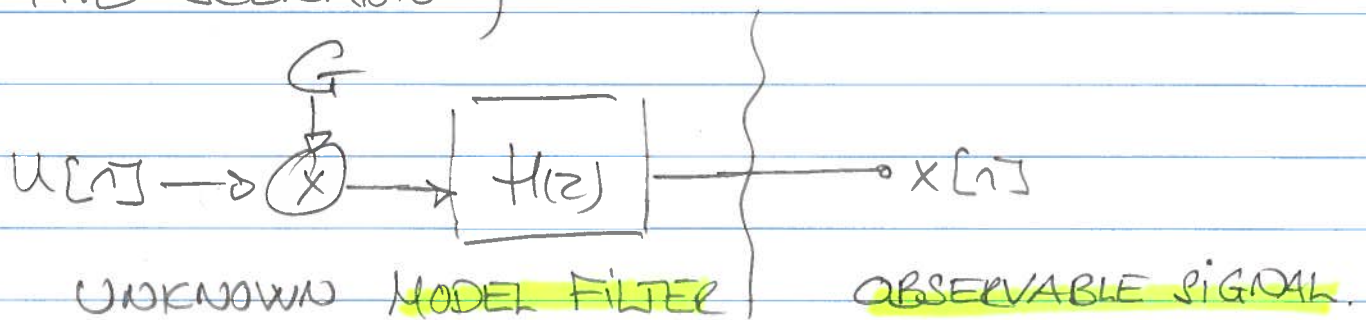
## Filter Design by Modelling

Assoc. Prof. Peter Koch, AAU

# ADVANCED SIGNAL PROCESSING, ES-7

## FILTER DESIGN BY MODELING — EXPLAINED FROM A FREQUENCY DOMAIN PERSPECTIVE.

FROM OUR PREVIOUS DISCUSSION, WE HAVE THE SCENARIO ;



ASSUMING THAT  $u[n]$  IS A WHITE SIGNAL AND  $x[n]$  IS AN AR PROCESS, WE HAVE THAT

$$\frac{X(z)}{U(z)} = G \cdot H(z) = \frac{G}{1 + \sum_{k=1}^p a_k z^{-k}}$$

WHERE  $p$  IS THE MODEL ORDER.

PREDICTING  $x[n]$  CAN BE DONE USING A LINEAR ONE-STEP PREDICTOR

$$\hat{x}[n] = - \sum_{k=1}^p \alpha_k x[n-k] \quad \text{AND THUS}$$

$$e[n] = x[n] - \hat{x}[n] \quad \text{IS THE PREDICTION ERROR (RESIDUAL)}$$

$$e[n] = x[n] - \hat{x}[n]$$

↓

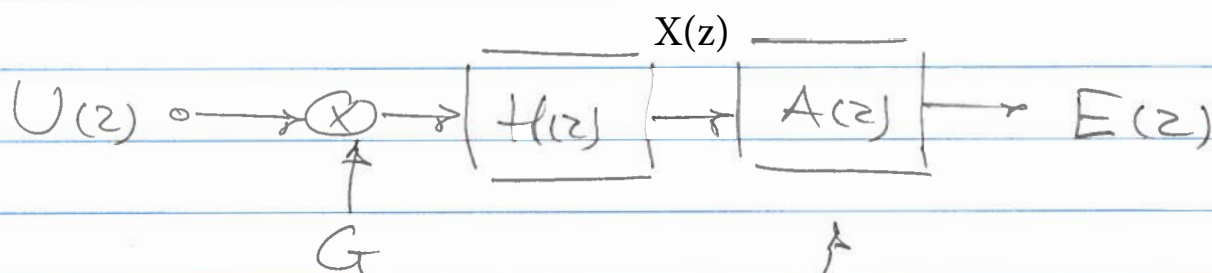
$$e[n] = x[n] + \sum_{k=1}^{p_1} \alpha_k x[n-k]$$

↓

$$E(z) = X(z) + \sum_{k=1}^{p_1} \alpha_k X(z) z^{-k}$$

↓

$$\frac{E(z)}{X(z)} = 1 + \sum_{k=1}^{p_1} \alpha_k z^{-k} = A(z)$$



THE INVERSE FILTER

o

IN ORDER TO FIND THE FILTER COEFFICIENT  $\alpha_k$ , AND THUS  $H(z) = 1/A(z)$ , WE CONDUCT A LEAST MEAN SQUARE OPTIMIZATION OF THE ERROR  $e[n]$ , i.e.,  $\min\{E\}$  WHERE  $E = \underline{e}^T \cdot \underline{e}$

• DIFFERENTIATE  $E$  W.R.T.  $\underline{a}$  AND SET  $E_{a_k} = 0$

OR

• USE THE ORTHOGONALITY PRINCIPLE

BOTH METHODS LEAD TO THE NORMAL EQUATIONS;

$$\boxed{\underline{X^T X a = X^T b}}$$

WHERE

$$X = \begin{bmatrix} 0 & 0 & \dots & \dots & \dots & 0 \\ x_0 & 0 & \dots & \dots & \dots & 0 \\ x_1 & x_0 & 0 & \dots & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_p & x_{p-1} & \dots & \dots & \dots & x_0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{N-1} & \dots & \dots & \dots & x_{N-1-p} & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & x_{N-1} & \vdots \end{bmatrix}$$

$$\underline{a} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{bmatrix} \quad \text{AND} \quad \underline{b} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_p \\ \vdots \\ x_{N-1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

HERE WE HAVE APPLIED A WINDOW FUNCTION TO  $x[n]$ , AND THUS THE NORMAL EQUATIONS ARE THE TULE-WALKER EQUATIONS (AUTO-CORR. METHOD)

IN THE AUTO-CORRELATION METHOD, THE YULE-WALKER EQUATIONS ARE ;

$$\underline{R} \underline{a} = -\underline{r}$$

WHERE

$$\underline{R} = \begin{bmatrix} R(0) & R(1) & R(2) & \dots & R(p-1) \\ R(1) & R(0) & R(1) & & \\ R(2) & R(1) & & & \\ \vdots & & & & \\ R(p-1) & & & & R(0) \end{bmatrix}$$

$$\underline{r} = \begin{bmatrix} R(1) \\ R(2) \\ \vdots \\ R(p) \end{bmatrix}$$

$\underline{R}$  IS A SYMMETRIC TOEPLITZ MATRIX, AND  
 = THUS THE SET OF  $p$  LINEAR EQUATIONS  
 WITH  $p$  UNKNOWN CAN BE SOLVED EFFICIENTLY  
 USING THE LEVINSON-DURBIN RECURSIVE ALG.

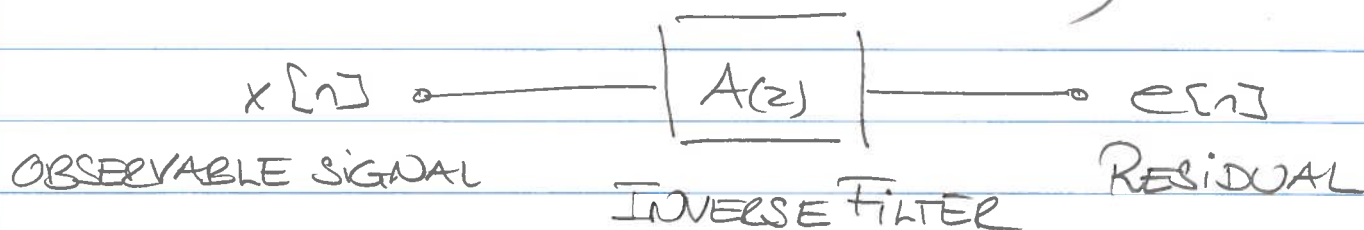
FURTHER WE FOUND THAT THE GAIN FACTOR  
 $G$ , FOR A GIVEN MODEL ORDER  $p$ , IS

$$G = \sqrt{E_{\min, p}}$$

WHERE

$$E_{\min, p} = R(0) + \sum_{k=1}^p \alpha_k \cdot R(k)$$

# FREQUENCY DOMAIN FORMULATION;



## RESIDUAL VARIANCE;

$$E_p = \sum_{n=-\infty}^{\infty} e^2[n] \quad (\text{AUTO-CORR. METHOD})$$

$$\text{VARIANCE} = \sum_{n=-\infty}^{\infty} y^2[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} |Y(e^{j\omega})|^2 d\omega$$

PARSEVAL IDENTITY

WHERE  $|Y(e^{j\omega})|^2$  IS THE POWER SPECTRUM

$$E_p = \frac{1}{2\pi} \int_{-\pi}^{\pi} |E(e^{j\omega})|^2 d\omega$$

The Inverse Filter

$$\begin{matrix} X(z) \\ X(e^{j\omega}) \end{matrix} \rightarrow \begin{matrix} A(z) \\ A(e^{j\omega}) \end{matrix} \rightarrow \begin{matrix} E(z) = A(z) \cdot X(z) \\ E(e^{j\omega}) = A(e^{j\omega}) \cdot X(e^{j\omega}) \end{matrix}$$

$$E_p = \frac{1}{2\pi} \int_{-\pi}^{\pi} |A(e^{j\omega}) \cdot X(e^{j\omega})|^2 d\omega$$

$$E_p = \frac{1}{2\pi} \int_{-\pi}^{\pi} |A(e^{j\omega})|^2 \cdot |X(e^{j\omega})|^2 d\omega$$

$$\Downarrow \quad E_p = \frac{1}{2\pi} \int_{-\pi}^{\pi} |A(e^{j\omega})|^2 \cdot |X(e^{j\omega})|^2 d\omega$$

$$E_p = \frac{1}{2\pi} \int_{-\pi}^{\pi} A(e^{j\omega}) \cdot A(e^{-j\omega}) \cdot P(\omega) d\omega \quad (\times 1)$$

WHERE  $P(\omega)$  IS THE POWER SPECTRUM OF  $X[n]$ ,  
AND  $A(e^{-j\omega})$  IS THE COMPLEX CONJUGATED OF  $A(e^{j\omega})$

Now, USE THIS EXPRESSION TO FIND THE  
MINIMAL RESIDUAL VARIANCE ;

$$\frac{\partial E_p}{\partial a_i} = 0 \quad \text{for } 0 \leq i \leq p$$

IN ORDER TO DO THIS, WE INTRODUCE SOME DEF. ;

$$A(e^{j\omega}) \triangleq [1 \ \underline{a}^T] \cdot \begin{bmatrix} 1 \\ \underline{\psi}(e^{j\omega}) \end{bmatrix} = [1 \ a_1 \ a_2 \ \dots \ a_p] \cdot \begin{bmatrix} 1 \\ e^{-j\omega} \\ e^{-j2\omega} \\ \vdots \\ e^{-jp\omega} \end{bmatrix} = (1 + \underline{a}^T \underline{\psi}(e^{j\omega}))$$

$$A(e^{-j\omega}) \triangleq [1 \ \underline{\psi}^T(e^{-j\omega})] \cdot \begin{bmatrix} 1 \\ \underline{a} \end{bmatrix} = [1 \ e^{j\omega} \ e^{j2\omega} \ \dots \ e^{jp\omega}] \cdot \begin{bmatrix} 1 \\ a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = (1 + \underline{\psi}^T(e^{-j\omega}) \underline{a})$$

PK

From (X1) p.6 ;

7.

$$A(e^{j\omega}) \cdot A(e^{-j\omega}) = (1 + \underline{a}^T \underline{\psi}(e^{j\omega})) (1 + \underline{\psi}^T(e^{-j\omega}) \cdot \underline{a})$$

$$= 1 + \underline{\psi}^T(e^{-j\omega}) \underline{a} + \underline{a}^T \underline{\psi}(e^{j\omega}) + \underline{a}^T \underline{\psi}(e^{j\omega}) \underline{\psi}^T(e^{-j\omega}) \underline{a}$$

THEREFORE ;

$$\frac{\partial E_p}{\partial a_i} = 0 \quad \text{for } 0 \leq i \leq p$$

⇓

$$\frac{\partial E_p}{\partial \underline{a}^T} = \frac{\partial}{\partial \underline{a}^T} \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} [A(e^{j\omega}) \cdot A(e^{-j\omega})] \cdot p(\omega) d\omega \right\} = \underline{0}$$

⇓

$$\frac{\partial E_p}{\partial \underline{a}^T} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\partial}{\partial \underline{a}^T} \left( 1 + \underline{\psi}^T(e^{-j\omega}) \underline{a} + \underline{a}^T \underline{\psi}(e^{j\omega}) + \underline{a}^T \underline{\psi}(e^{j\omega}) \underline{\psi}^T(e^{-j\omega}) \underline{a} \right) \cdot p(\omega) d\omega = \underline{0}$$

⇓

$$\frac{\partial E_p}{\partial \underline{a}^T} = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[ \underline{\psi}^T(e^{-j\omega}) + \underline{\psi}^T(e^{-j\omega}) \underline{\psi}(e^{j\omega}) \underline{\psi}^T(e^{-j\omega}) \underline{a} + \underline{\psi}^T(e^{-j\omega}) \underline{\psi}(e^{j\omega}) \underline{\psi}^T(e^{-j\omega}) \underline{a} \right] \cdot p(\omega) d\omega = \underline{0}$$

LET'S DENOTE THIS MATRIX M

$$\underline{M} = \underline{\Psi}(e^{j\omega}) \cdot \underline{\Psi}^T(e^{-j\omega})$$

$$= \begin{bmatrix} e^{-j\omega} \\ e^{-j2\omega} \\ \vdots \\ e^{-jp\omega} \end{bmatrix} \cdot \begin{bmatrix} e^{j\omega} & e^{j2\omega} & \dots & e^{jp\omega} \end{bmatrix}$$

$$= \begin{bmatrix} e^0 & e^{j\omega} & e^{j2\omega} & \dots & e^{j(p-1)\omega} \\ e^{-j\omega} & e^0 & e^{j\omega} & \dots & e^{j(p-2)\omega} \\ e^{-j2\omega} & e^{-j\omega} & e^0 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ e^{-j(p-1)\omega} & \dots & \dots & \dots & e^0 \end{bmatrix}$$

AND THEREFORE ;

$$\underline{\Psi}(e^{-j\omega}) \underline{\Psi}^T(e^{j\omega}) = \begin{bmatrix} e^{j\omega} \\ e^{j2\omega} \\ \vdots \\ e^{jp\omega} \end{bmatrix} \begin{bmatrix} e^{-j\omega} & e^{-j2\omega} & \dots & e^{-jp\omega} \end{bmatrix}$$

$$= \underline{M}^T$$

PK.

9.

LET'S AGAIN LOOK AT THE INTEGRAL

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \left[ \underline{\Psi}(e^{-j\omega}) + \underline{\Psi}(e^{j\omega}) + \underline{\Psi}(e^{j\omega}) \underline{\Psi}^T(e^{-j\omega}) \underline{a} + \underline{\Psi}(e^{-j\omega}) \underline{\Psi}^T(e^{j\omega}) \underline{a} \right] P(\omega) d\omega$$

= 0

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[ \underline{\Psi}(e^{-j\omega}) + \underline{\Psi}(e^{j\omega}) + \underline{M} \underline{a} + \underline{M}^T \underline{a} \right] P(\omega) d\omega = \underline{0}$$



$$(\underline{M} + \underline{M}^T) \underline{a}$$

$$\begin{bmatrix} e^{j\omega} \\ e^{j2\omega} \\ \vdots \\ e^{jp\omega} \end{bmatrix} + \begin{bmatrix} e^{-j\omega} \\ e^{-j2\omega} \\ \vdots \\ e^{-jp\omega} \end{bmatrix}$$

$$= \begin{bmatrix} \cos \omega + j \sin \omega \\ \cos 2\omega + j \sin 2\omega \\ \vdots \\ \cos p\omega + j \sin p\omega \end{bmatrix} + \begin{bmatrix} \cos \omega - j \sin \omega \\ \cos 2\omega - j \sin 2\omega \\ \vdots \\ \cos p\omega - j \sin p\omega \end{bmatrix} = 2 \begin{bmatrix} \cos \omega \\ \cos 2\omega \\ \vdots \\ \cos p\omega \end{bmatrix}$$

$$\underline{M} + \underline{M}^T = 2 \cdot \begin{bmatrix} 1 & \cos \omega & \cos 2\omega & \dots & \cos((p-1)\omega) \\ \cos \omega & 1 & \cos \omega & \dots & \cos((p-2)\omega) \\ \cos 2\omega & \cos \omega & 1 & & \\ \vdots & & & \ddots & \\ \cos((p-1)\omega) & \cos((p-2)\omega) & \dots & & 1 \end{bmatrix}$$

PK.

10.

AND THUS ;

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \left\{ 2 \begin{bmatrix} \cos \omega \\ \cos 2\omega \\ \vdots \\ \cos p\omega \end{bmatrix} + 2 \begin{bmatrix} 1 & \cos \omega & \dots & \cos(p-1)\omega \\ \cos \omega & 1 & & \\ \vdots & & \ddots & \\ \cos(p-1)\omega & & & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} \right\} P(\omega) d\omega = \underline{\underline{0}}$$

SO FAR SO GOOD...

NOW, LET'S HAVE A LOOK AT  $P(\omega)$  :

$$P(\omega) = |X(e^{j\omega})|^2$$

$$\Downarrow P(\omega) = X(e^{j\omega}) X(e^{-j\omega})$$

CONDUCTING AN INVERSE DISCRETE-TIME FT ;

$$\Downarrow \text{IDTFT} \{ P(\omega) \} = \text{IDTFT} \{ X(e^{j\omega}) \cdot X(e^{-j\omega}) \}$$

$$\Downarrow p[n] = x[n] * x[-n]$$

$$\Downarrow p[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot x[n+k]$$

$$\Downarrow p[n] = \sum_{k=0}^{N-1} x[k] x[n+k]$$

Auto-Corr.  
METHOD

$$\Downarrow p[n] = R(n)$$

↑ THE  $n^{\text{th}}$  AUTO-CORR. LAG.

PK

So ;  $R(n) \xleftrightarrow{\text{DTFT}} P(\omega)$  11

$$R(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(\omega) \cdot e^{j\omega n} d\omega$$

$$R(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(\omega) \{ \cos(\omega n) + j \sin(\omega n) \} d\omega$$

THE POWER SPECTRUM  $P(\omega)$  is REAL  
EVEN

$$R(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(\omega) \cdot \cos(\omega n) d\omega$$

WE NOW COMPARE THIS TO THE INTEGRAL ;

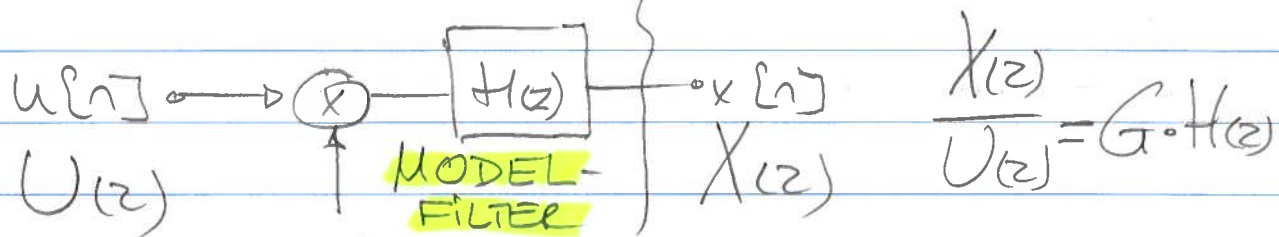
$$\frac{1}{2\pi} \int_{-\pi}^{\pi} 2 \left\{ \begin{bmatrix} \cos \omega \\ \cos 2\omega \\ \vdots \\ \cos p\omega \end{bmatrix} + \begin{bmatrix} 1 & \cos \omega & \dots & \cos(p-1)\omega \\ \cos \omega & 1 & & \\ \vdots & & \ddots & \\ \cos(p-1)\omega & \dots & & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} \right\} P(\omega) d\omega = 0$$

$$2 \cdot \begin{bmatrix} R(1) \\ R(2) \\ \vdots \\ R(p) \end{bmatrix} + 2 \cdot \begin{bmatrix} R(0) & R(1) & \dots & R(p-1) \\ R(1) & R(0) & & \\ \vdots & & \ddots & \\ R(p-1) & \dots & & R(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = 0$$

$$\underline{R} \underline{a} = -\underline{r}$$

TOLE-WALKER EQUATIONS

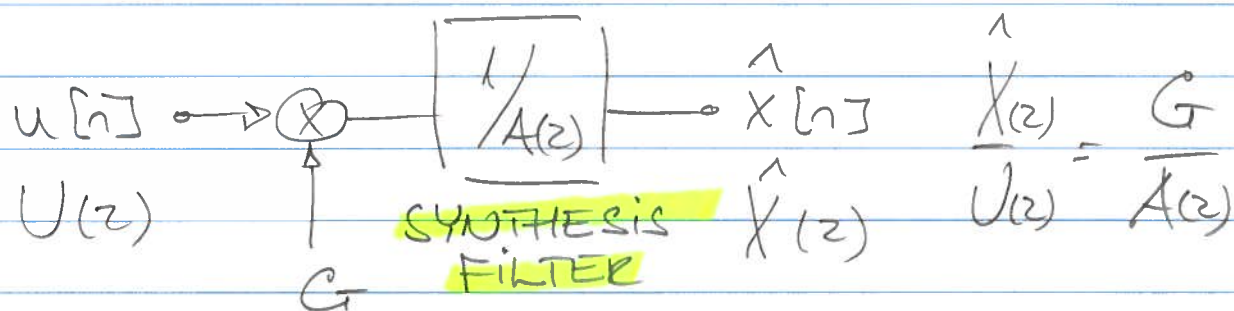
How is  $P(\omega)$  APPROACHED ?



$$|G \cdot H(e^{j\omega})|^2 = \frac{|X(e^{j\omega})|^2}{|U(e^{j\omega})|^2} = \frac{|X(e^{j\omega})|^2}{|U(e^{j\omega})|^2} = P(\omega)$$

BECAUSE  $|X(e^{j\omega})|^2 = P(\omega)$  AND  $u[n]$  IS WHITE

$$\Downarrow \quad P(\omega) = G^2 \cdot |H(e^{j\omega})|^2$$

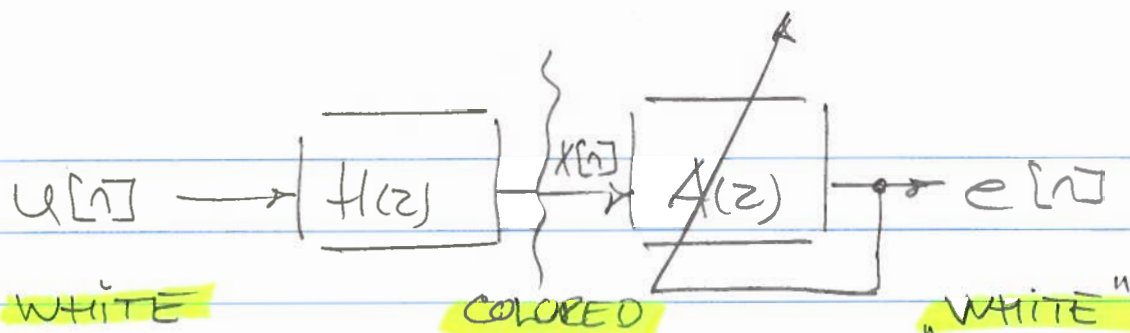


$$\left| \frac{G}{A(e^{j\omega})} \right|^2 = \frac{|\hat{X}(e^{j\omega})|^2}{|U(e^{j\omega})|^2} = \frac{|\hat{X}(e^{j\omega})|^2}{|U(e^{j\omega})|^2} = \hat{P}(e^{j\omega})$$

BECAUSE  $|\hat{X}(e^{j\omega})|^2 = \hat{P}(e^{j\omega})$  AND  $u[n]$  IS WHITE.

$$\Downarrow \quad \hat{P}(\omega) = G^2 \cdot \frac{1}{|A(e^{j\omega})|^2}$$





THEREFORE,  $A(z)$  IS THE FILTER WHICH IN THE WUS-SENSE MAKES  $e[n]$  A WHITE SIGNAL.

$A(z)$  IS THUS DENOTED AS WHITENING FILTER

Now, since  $\hat{P}(\omega) = \frac{G^2}{|A(e^{j\omega})|^2} = \frac{G^2}{|1 + \sum_{k=1}^p a_k e^{-j\omega k}|^2}$

THEN WE CAN CONCLUDE THAT ANY SPECTRUM  $P(\omega)$  CAN BE APPROXIMATED BY AN AR-MODEL, SUBJECT TO A SUFFICIENTLY LARGE MODEL ORDER  $p$

IT CAN BE SHOWN THAT

$$\hat{P}(\omega) = \frac{G^2}{g(0) + 2 \sum_{i=1}^p g(i) \cos(i\omega)}$$

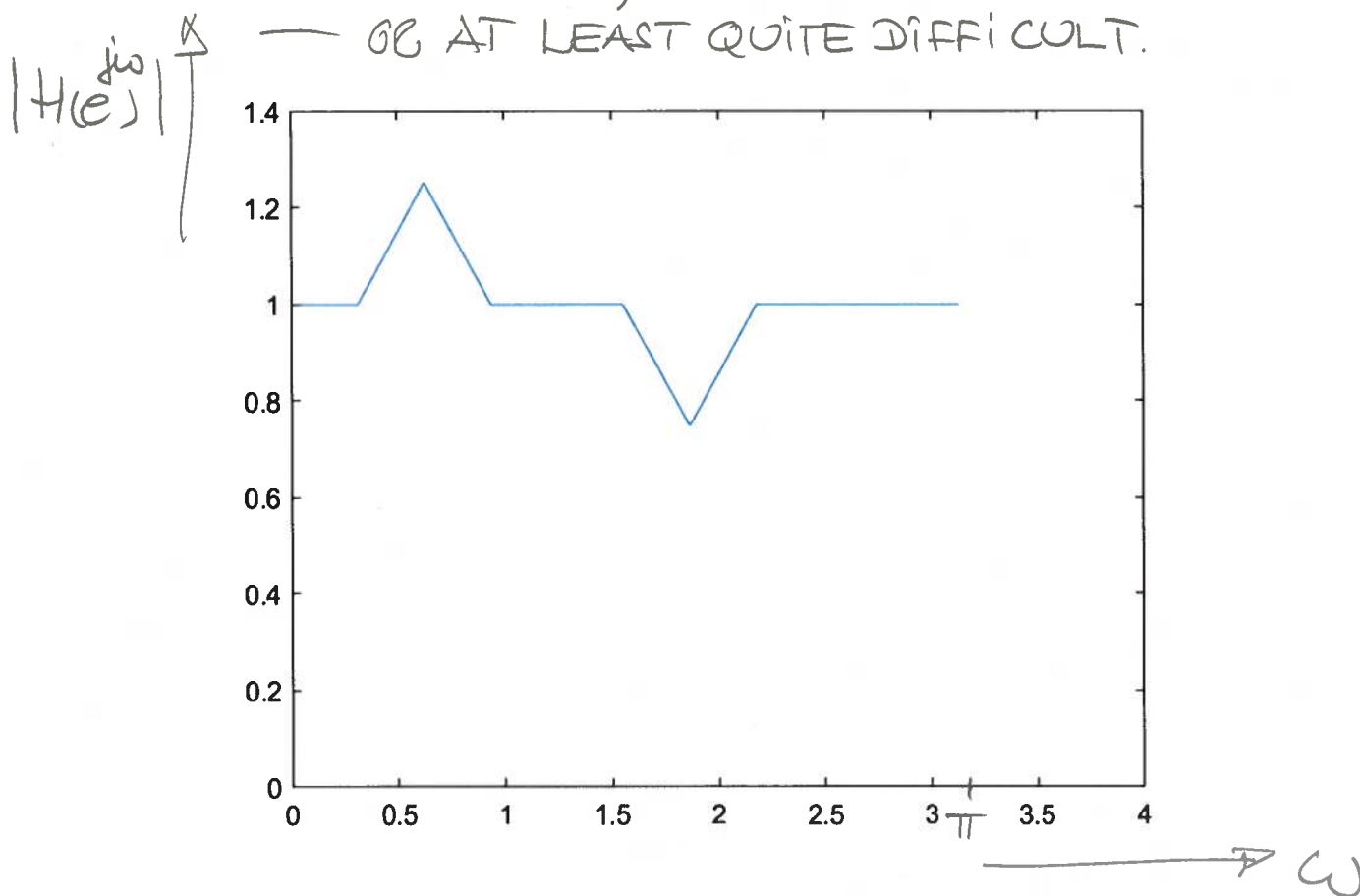
WHERE  $g(i) = \sum_{k=0}^{p-i} a_k a_{k+i} \quad 0 \leq i \leq p$

IS THE AUTO-CORRELATION OF THE IMPULSE RESPONSE OF THE INVERSE FILTER

LET'S ASSUME THAT WE WANT TO DESIGN AN IIR (AE) FILTER WITH THE FOLLOWING AMPLITUDE SPECTRUM,

USING "ORDINARY" METHODS (WHICH TAKES AN OUTSET IN A CONTINUOUS-TIME TRANSFER FUNCTION  $H(s)$ ) THIS SEEMS IMPOSSIBLE.

— OR AT LEAST QUITE DIFFICULT.



HOWEVER, USING LPC-BASED DESIGN, WE MIGHT HAVE A CHANCE...

So, HERE IS THE OVERALL IDEA ;

FROM p. 11 WE HAVE THAT

$$R(n) \xleftrightarrow{\text{DTFT}} P(\omega)$$

HEREFORE, KNOWING THE <sup>AMPLITUDE</sup> DESIRED SPECTRUM  $|H(e^{j\omega})|$ , WE CAN DERIVE THE POWER SPECTRUM  $|H(e^{j\omega})|^2$  AND THEN CALCULATE THE AUTO-CORRELATION COEFFICIENTS ;

$$R(n) = \text{IDTFT} \left\{ |H(e^{j\omega})|^2 \right\}$$

NEXT,  $R(n)$  CAN BE APPLIED IN THE YULE-WALKER EQUATIONS ;

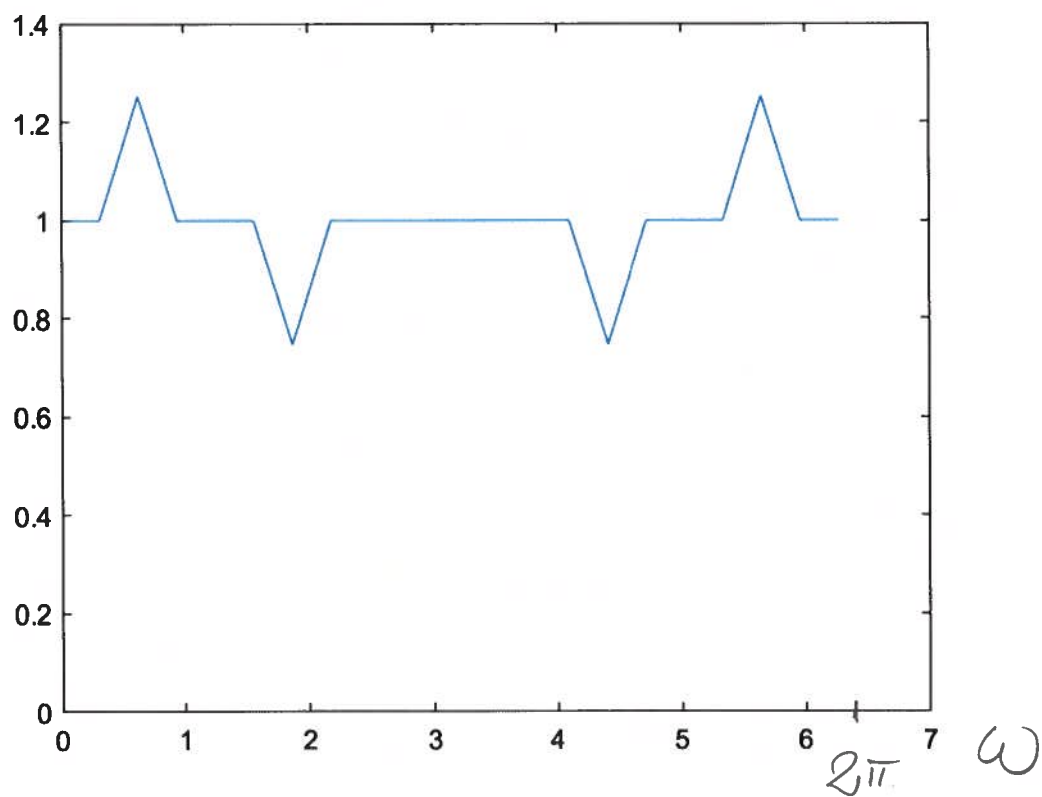
$$\underline{R} \underline{a} = -\underline{r} \Rightarrow A(z)$$

AND THEN  $\hat{P}(\omega) = \left| \frac{G}{A(e^{j\omega})} \right|^2$

$$= \frac{G^2}{\left| 1 + \sum_{k=1}^P a_k e^{-jk\omega} \right|^2}$$

First we extend the spectrum to  $2\pi$

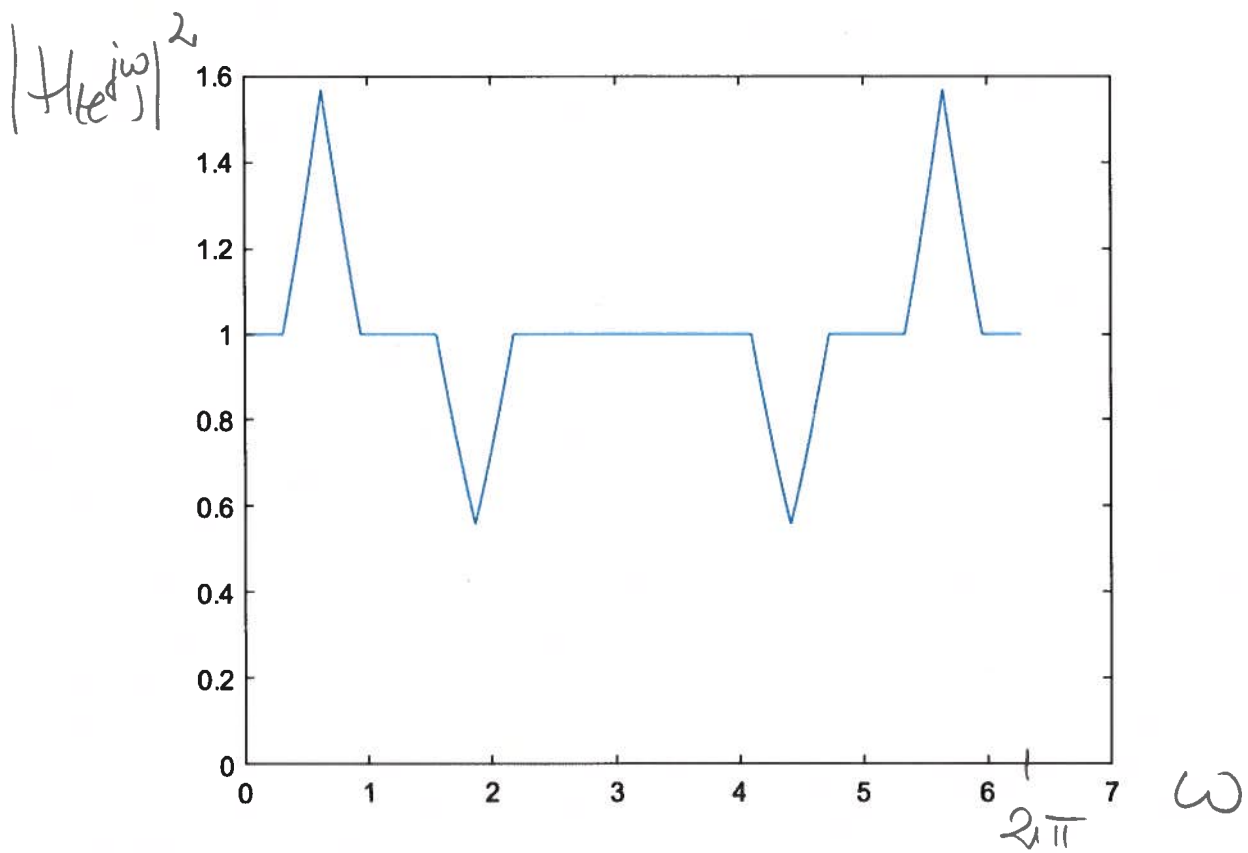
$|H|$



PK.

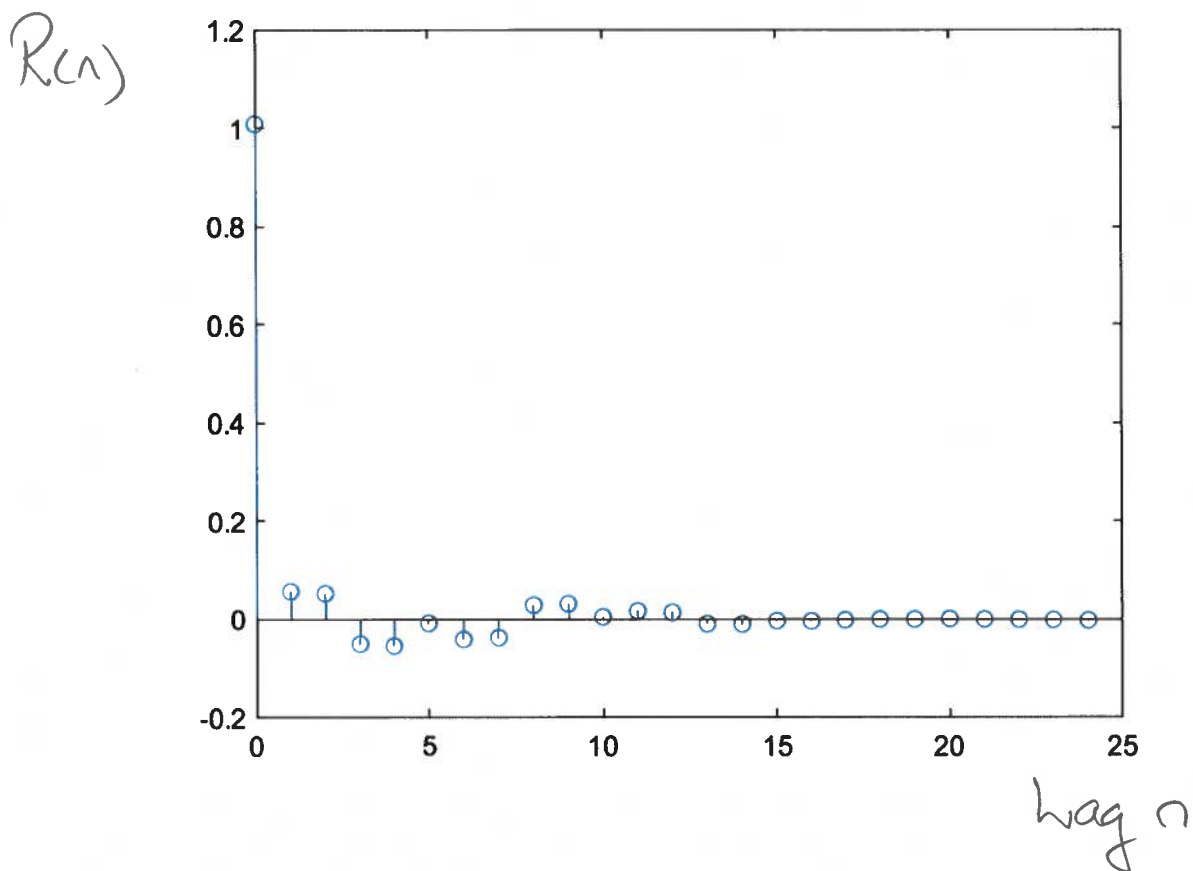
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THEN CONVERT TO POWER SPECTRUM

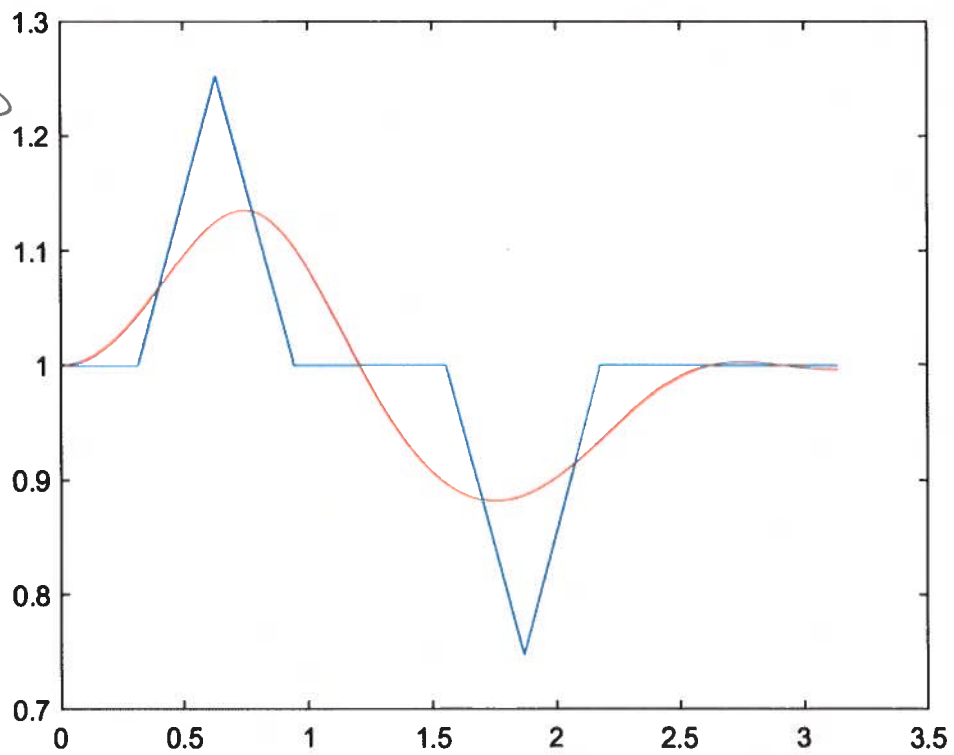


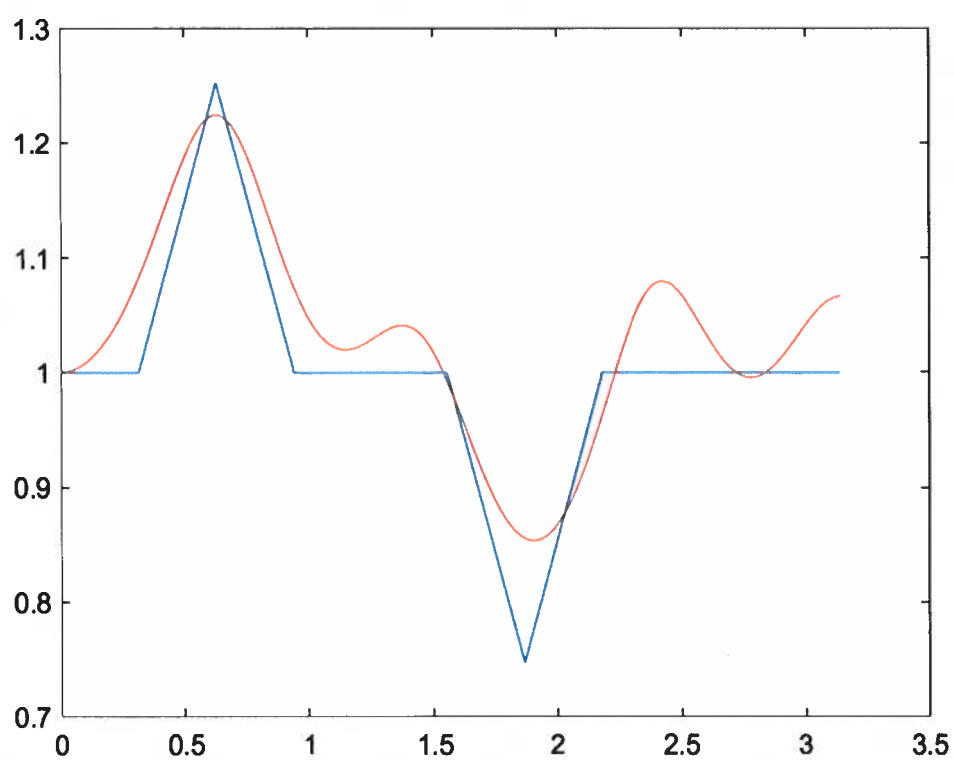
$$R(n) = \text{IDTFT} \{ |H(e^{j\omega})|^2 \}$$

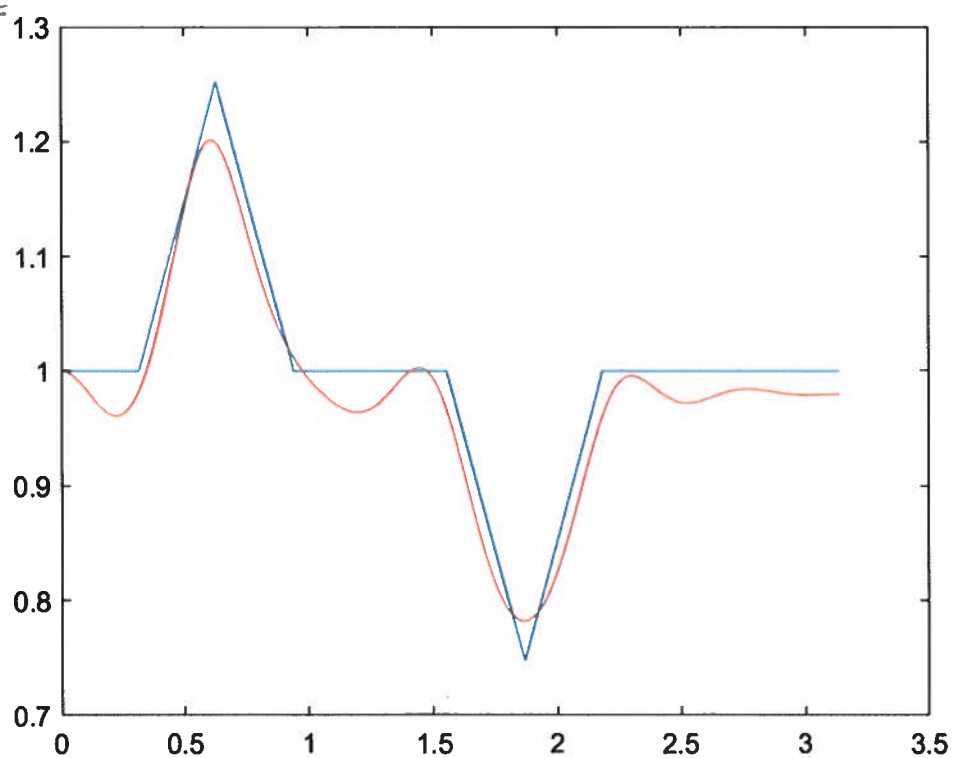
(CALCULATED USING IFFT)



NEXT BEING USED IN  $R_a = -r$

AR-FILTER,  $p=4$  $|H|$  BLUE $|\frac{1}{A}|$  RED $\omega$

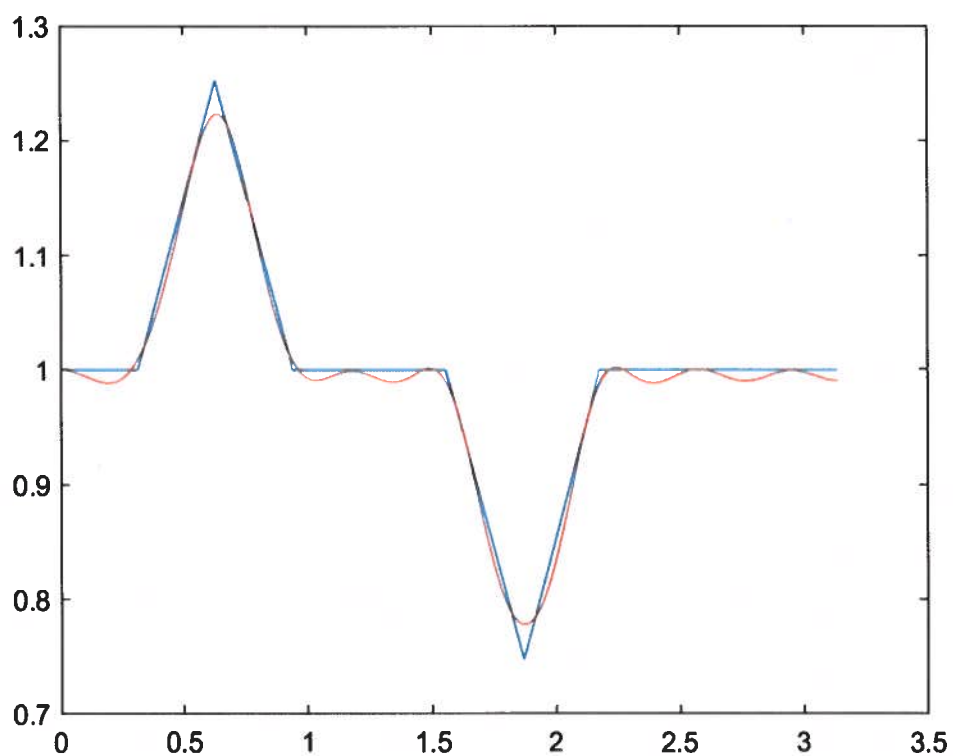
AR-FILTER,  $\rho=8$  $|H|$  BLUE $|1/A|$  RED $\omega$

AR-FILTER,  $p=12$  $|H|$  BLUE $|A|$  RED $\omega$

AR-FILTER,  $\rho = 16$

$|H|$  BLUE

$|1/A|$  RED



W

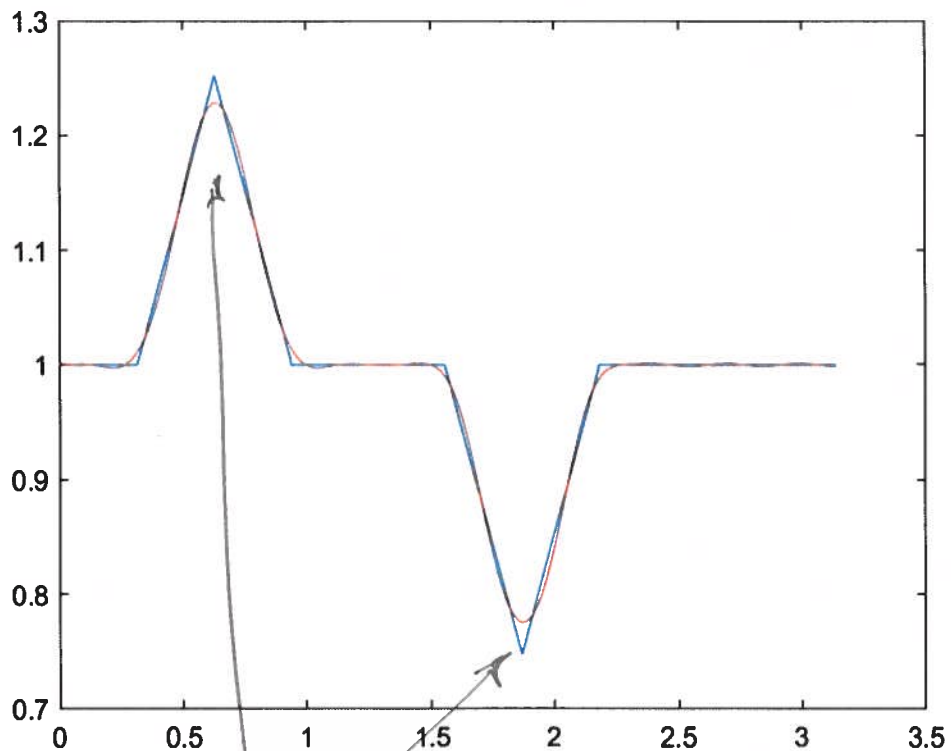
PK.

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AR-FILTER,  $p=20$

$|H|$  BLUE

$|\frac{1}{A}|$  RED



CAN YOU SPOT THE DIFFERENCE ?

$$1 - \beta z^{-1} = \frac{1}{1 + \alpha_1 z^{-1} + \alpha_2 z^{-2} + \dots + \alpha_\infty z^{-\infty}}$$